PRESENTATIONS & FINAL EVALUATION

1. Prepare a 10-slide presentation that explains your IT infrastructure project, including thetechnical setup, business problem it solves, and key features. Use real-world examples from apps like Zomato, Flipkart, or Paytm to illustrate your points.

Answer & Presentation Outline (10 Slides):

- **Slide 1:** Title Slide (Co-Working Space Network Architecture & Infrastructure Design)
- **Slide 2:** Business Problem (Addressing unsegmented Wi-Fi, security vulnerabilities, and network congestion in modern shared offices)
- **Slide 3:** Real-World Analogy (Comparing office departmental isolation to how Zomato separates delivery dispatch, customer support, and financial merchant portals)
- **Slide 4:** Proposed Technical Setup (Hierarchical star topology with core firewall and managed switches)
- **Slide 5:** VLAN Segmentation & IP Addressing (Isolating Marketing, Tech, Finance, and Guest VLAN 99 subnets)
- **Slide 6:** Network Redundancy & High Availability (Dual core switches with Spanning Tree Protocol and link aggregation)
- **Slide 7:** Hardware Stack & Bill of Materials (Fortinet firewall, TP-Link managed switches, and WiFi 6 access points under ₹1,00,000)
- **Slide 8:** Security Policy Framework (WPA3-Enterprise authentication, inter-VLAN blocking, and default-deny firewall rules)
- **Slide 9:** Scalability & Future Expansion (Transitioning from static routing to OSPF upon scaling)
- **Slide 10:** Conclusion & Q&A

2. Write a one-page technical summary document for your project, clearly describing themain components, technologies used, and how they work together to deliver the solution.

Answer & One-Page Technical Summary:

Project Title: Co-Working Space Enterprise Network Architecture

Executive Overview: This project delivers a secure, high-performance, and cost-effective IT network designed for a multi-department co-working space. The architecture integrates advanced Layer 2/Layer 3 networking principles to ensure seamless data communication and robust security.

Core Components & Technologies:

- **Gateway & Security:** Fortinet FortiGate 40F Next-Gen Firewall providing stateful inspection, NAT, and default-deny inter-VLAN routing rules.
- **Switching Infrastructure:** TP-Link JetStream L2+ Managed Switches utilizing 802.1Q VLAN trunking and Spanning Tree Protocol (STP) for loop-free redundancy.

- **Wireless Access:** Wi-Fi 6 Access Points (EAP650) supporting WPA3 encryption, fast roaming, and isolated guest portals.

Operational Flow: Incoming ISP traffic hits the firewall gateway, where security policies inspect packets before routing them across department-specific VLANs (Marketing, Tech, Finance). Access switches distribute traffic locally, while redundant uplinks guarantee zero downtime during device failures.

3. List three questions you think the evaluation panel might ask about your project'stechnical correctness or business alignment, and write a brief answer for each.

Answer & Evaluation Panel Q&A:

- **Q1 (Security):** "How does your design prevent a compromised guest device from accessing sensitive financial databases?"
  Answer: Guest traffic is segregated onto VLAN 99 with Layer 2 client isolation enabled. Furthermore, inter-VLAN routing rules on the firewall block all direct traffic between guest networks and internal corporate subnets.
- **Q2 (Redundancy):** "What happens to office connectivity if the primary core switch encounters a power failure?"
  Answer: Our redundancy plan deploys dual core switches connected via Spanning Tree Protocol (STP) and redundant trunk links, allowing traffic to automatically failover to secondary paths within seconds.
- **Q3 (Business Alignment):** "How does this network architecture support the financial goals of a startup co-working space?"
  Answer: By utilizing enterprise-grade managed gear from cost-effective brands like TP-Link and MikroTik/Fortinet, we achieve high reliability and security under a strict budget of under ₹1,00,000 INR, avoiding expensive enterprise licensing fees.

4. Review your project documentation and checklist to ensure every required feature isimplemented and documented. Mark any incomplete sections and write a short plan for how you would finish them if given more time.

Answer & Audit Plan:

Audit Status: All core deliverables—including network topologies, IP subnets, Bill of Materials, security policies, and peer review updates—are fully completed and documented.

Future Enhancement Plan (If More Time Allowed):

- 1. Implement 802.1X Network Access Control (NAC) via a RADIUS server for automated device profiling.
- 2. Configure automated SNMP network monitoring using Zabbix or PRTG for real-time bandwidth and uptime tracking.

5. After your presentation, write a reflection (5-7 sentences) describing the feedback youreceived from the panel and what specific improvements you would make to your project or presentation skills based on that feedback.

Answer & Project Reflection:

The evaluation panel appreciated the thoroughness of our security policy framework and the practical cost-benefit justification of our Bill of Materials. However, they suggested that we include a clearer visual breakdown of how DHCP relay operates across multiple remote subnets. In response, I would add a dedicated packet flow diagram illustrating DHCP discover-offer-request-ack transactions across VLAN boundaries. Overall, the presentation boosted my confidence in defending technical architecture choices before an expert panel. I also learned the importance of balancing high-level business value with deep-dive technical configurations during live pitches. This experience has significantly strengthened my understanding of enterprise network design and professional project presentation.